SMARTWASTE

Smart Solutions for a Cleaner Urban Future

PROJECT REPORT

SEVA First Innovation Challenge
Theme: Urban Development
Focus: Solid Waste Management, Waste Segregation,
Circular Economy and Construction Waste

Submitted by
Team GreenLoop Innovators

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Academic Project / Prototype Report

1. Executive Summary

SmartWaste is a frontend prototype concept designed to encourage responsible waste segregation and improve awareness of solid waste management in urban communities. It provides a simple digital interface for reviewing waste categories, learning how to dispose of common materials, and recording sample waste-related reports in the browser. The project promotes reduce, reuse, recycle and responsible construction-waste handling. The current version is a demonstration prototype and is not connected to a municipal waste-management system.

2. Introduction

Rapid urbanisation increases the amount and variety of waste generated by households, shops, institutions and construction activities. Mixed or improperly disposed waste can create unhygienic surroundings, block drains, increase landfill pressure and make recycling more difficult. SmartWaste proposes a user-friendly digital awareness platform to help people understand waste segregation and make more responsible disposal decisions.

3. Problem Statement

Communities face challenges such as mixing wet and dry waste, uncertainty about where materials belong, limited recycling awareness, and improper handling of construction and demolition waste. These practices reduce the quality of recyclable materials and can contribute to pollution. Accessible information and simple reporting concepts can encourage better waste-management habits.

4. Proposed Solution

SmartWaste provides a simple web interface that explains waste categories and segregation practices. The prototype includes a waste-report form for entering sample details and a dashboard-style area for viewing entries stored in the same browser. It demonstrates how a digital platform could support awareness, community participation and more organised waste reporting in a future version.

- Provide easy-to-understand guidance for separating common waste types.
- Encourage users to report waste-related concerns through a simple interface.
- Demonstrate a basic dashboard for viewing locally saved sample reports.
- Promote recycling, reuse and responsible construction-waste handling.
- Create a foundation for future integration with verified local services.

5. Objectives

1. Increase awareness about wet, dry, recyclable, hazardous and construction waste.
2. Encourage correct segregation before disposal.
3. Demonstrate a simple digital reporting workflow.
4. Support community participation in cleaner urban spaces.
5. Introduce circular-economy thinking by promoting reduction, reuse and recycling.
6. Identify future improvements required for a reliable real-world system.

6. Main Features of the Prototype

6.1 Waste Segregation Guide

The guide explains that waste should be separated according to local municipal rules. Examples include food scraps and garden waste as wet/organic waste; clean paper, cardboard, metal and certain plastics as dry/recyclable waste; batteries, chemicals and electronic waste as special-handling items; and bricks, concrete, tiles and rubble as construction and demolition waste. Exact collection rules can differ by locality.

6.2 Waste Reporting Form

Users can enter sample details about a waste concern, such as a category and a short description. This demonstrates the information a future reporting system could collect. In the current prototype, entries remain in the browser and are not sent to any authority.

6.3 Local Dashboard / Saved Reports

The interface can display reports saved locally in the same browser using localStorage. These records are browser-specific and may not appear on another device or browser. This is a demonstration feature, not a shared live database.

6.4 Circular Economy Awareness

The project promotes reducing unnecessary consumption, reusing items when practical, recycling suitable materials and ensuring special or construction waste is handled through appropriate local channels.

7. How the Prototype Works

7. The user opens the SmartWaste webpage in a web browser.
8. The user reviews the waste categories and segregation guidance.
9. The user enters sample information in the reporting form.
10. The webpage processes the form in the browser.
11. The sample entry is saved locally in the browser, where supported.
12. The user can view locally stored entries in the prototype interface.

Note: The current prototype does not provide real-time waste-collection tracking, automatic pickup scheduling, server-side image uploads, municipal integration, or cross-device data synchronisation.

8. Technology Stack

- HTML: structures the webpage and its content.
- CSS: controls layout, styling, spacing and responsive presentation.
- JavaScript: handles browser-side interactions and form behaviour.
- Browser localStorage: stores demonstration entries locally in the user's browser.

9. Waste Segregation Guidelines

These are general examples. Users should follow local municipal collection rules.

- Wet/organic waste: fruit and vegetable peels, leftover food and garden waste where accepted.
- Dry recyclable waste: clean paper, cardboard, metal cans, glass and recyclable plastics accepted by local services.

- Domestic hazardous waste: batteries, bulbs, paints, chemicals and similar items should be kept separate and taken to approved collection points.
- E-waste: old phones, chargers, computers and electronic devices should go to authorised e-waste collection or recycling channels.
- Construction and demolition waste: bricks, concrete, tiles, rubble and similar materials should be kept separate from household waste and sent through approved local services.

10. Expected Benefits

- Improved public understanding of waste categories and segregation.
- More awareness of recycling and circular-economy practices.
- A simple interface to demonstrate community reporting.
- Potential to reduce mixed waste if adopted alongside suitable local services.
- A starting point for a more complete waste-management platform.

11. Limitations

- The current version is a frontend demonstration prototype.
- Reports are stored locally in the browser and are not shared with other users.
- There is no backend server or central database.
- There is no integration with municipal authorities, collection vehicles or recycling centres.
- The prototype does not verify reports or provide real-time status updates.
- Any photo field or file selector, if shown, does not mean a photo is uploaded to a server.

12. Future Scope

- Add a secure backend and database for shared report storage.
- Introduce user accounts, authentication and role-based access.
- Integrate with municipal departments only after formal coordination and permission.
- Provide verified collection-centre information and local disposal instructions.
- Add image uploads, location tagging and report-status notifications with suitable privacy controls.
- Explore route optimisation and collection scheduling after obtaining reliable operational data.
- Measure user engagement and segregation outcomes through a pilot study.

13. Conclusion

SmartWaste demonstrates how a simple web interface can communicate waste-segregation guidance and illustrate a local waste-reporting workflow. It highlights the importance of community awareness, recycling, reuse and responsible construction-waste handling. Although the present version is only a frontend concept prototype, it provides a basis for further development, testing and collaboration with local waste-management stakeholders.

14. Team Details

Team Member	Role	Main Contribution
Bhagyashri Pawar	Team Leader & Research	Coordination, research and project direction

Anjali Tir amal	Technical Development	Frontend implementation and technical work
Shraddha Rajput	UI/UX Design	Interface layout and user experience
Rohini Saindane	Data Collection & Analysis	Collecting information and organising findings
Yashasri Sapkale	Presentation & Documentation	Report preparation and presentation

Prepared for the SEVA First Innovation Challenge. This report describes the current concept prototype and its intended future direction.